

Name: _____



Investigating transformations with GeoGebra

Task A: Using GeoGebra for rotations and enlargements

1) [Task A Question 1](#)

Enlarge object A onto image A'.

- a) What is the scale factor?
- b) At what coordinate is the centre of enlargement?

2) [Task A Question 2](#)

Enlarge object B onto image B'.

- a) What is the scale factor?
- b) At what coordinate is the centre of enlargement?

3) [Task A Question 3](#)

Rotate object C onto image C'.

- a) What is the angle of rotation?
- b) At what coordinate is the centre of rotation?

4) [Task A Question 4](#)

Use the slider to rotate object D onto image D'.

- a) What is the angle of rotation?



Task B: Using GeoGebra for reflections and translations

1) [Task B Question 1](#)

Draw a line through points P and Q.

Reflect object A over the line.

Move the line of reflection down onto each of the numbered lines.

Which line reflects object A onto image A'?

2) [Task B Question 2](#)

Reflect object B over the line.

Move point P to each of the number points to turn the line.

At which point does the line reflect object B onto image B'?

3) [Task B Question 3](#)

Translate each of the objects A to E using vectors 1 to 9.

Which vector translates each object onto the image?

- a) Vector translates object A onto the image.
- b) Vector translates object B onto the image.
- c) Vector translates object C onto the image.
- d) Vector translates object D onto the image.
- e) Vector translates object E onto the image.

Task C: Using GeoGebra to identify transformations

[Task C](#)

Find and describe each transformation below.

- 1. $A \mapsto B$
- 2. $A \mapsto C$
- 3. $A \mapsto D$
- 4. $A \mapsto E$